

# Practical 1

## Practice Lab Assignment

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1.1.1. Calculate Momentum

100%

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum  $p$  is calculated using the formula:

$$p = m \times v$$

where:

- $m$  is the mass of the object (in kilograms).
- $v$  is the velocity of the object (in meters per second).

**Input Format:**

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

**Output Format:**

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 m=float(input())
2 v=float(input())
3 p=m*v
4 print('%.2f'%p,end='')
5 print("kgm/s")
```

Average time0.010 s10.25 ms

Maximum time0.014 s14.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 114 ms

Expected output5.010.050.00kgm/s

Actual output5.010.050.00kgm/s

Test case 27 ms

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1.1.2. Conditional Calculation Based on the Number of Digits

57%

Write a Python program that accepts an integer  $n$  as input. Depending on the number of digits in  $n$ .

**Constraints:**

 $1 \leq n \leq 999$ 

**Input Format:**

- The input consists of a single integer  $n$ .

**Output Format:**

- If  $n$  is a single-digit number, print its square.
- If  $n$  is a two-digit number, print its square root (rounded to two decimal places).
- If  $n$  is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 n=int(input())
2 if (0<=n<10):
3     print(n**2)
4 elif(10<=n<100):
5     a=n**(1/2)
6     print('%.2f'%a)
7 elif (100<=n<1000):
8     b=n**(1/3)
```

Average time0.004 s3.57 ms

Maximum time0.006 s6.00 ms

4 out of 4 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 16 ms

Expected output981

Actual output981

Test case 23 ms

Test case 34 ms

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1.1.3. Days between Two Dates

14/22

Write a Python program to calculate number of days between two dates.

**Input Format:**

- The first line contains the first date in the format  $YYYY-MM-DD$ .
- The second line contains the second date in the format  $YYYY-MM-DD$ .

**Output Format:**

- An integer representing the number of days between the given dates.

**Note:**

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

```
1 from datetime import date
2 def days_between_dates(date_str1, date_str2):
3     # Parse the first date string
4     year1, month1, day1 = map(int, date_str1.split('-'))
5     date1 = date(year1, month1, day1)
6
7     # Parse the second date string
8     year2, month2, day2 = map(int, date_str2.split('-'))
```

Average time0.006 s6.00 ms

Maximum time0.008 s8.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 18 ms

Expected output2014-07-022014-11-02123

Actual output2014-07-022014-11-02123

Test case 5 ms

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### 1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

**Input Format:**

- The input is a single integer.

**Output Format:**

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```

1 n=int(input())
2 rev=int(str(n)[::-1])
3 print(rev)

```

Average time  
0.003 s  
3.40 ms

Maximum time  
0.004 s  
4.00 ms

2 out of 2 shown test case(s) passed  
3 out of 3 hidden test case(s) passed

Test case 1

Expected output

5367

7635

Actual output

5367

7635

Test case 2

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### 1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

**Input Format:**

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

**Output Format:**

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

**Note:**

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```

1 #write your code here...
2 n=int(input())
3 for i in range(1,11):
4     print(f"{n} x {i} = {n*i}")
5
6

```

Average time  
0.004 s  
4.00 ms

Maximum time  
0.005 s  
5.00 ms

2 out of 2 shown test case(s) passed  
2 out of 2 hidden test case(s) passed

Test case 1

Expected output

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

Actual output

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

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# Lab Assignment

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1.2.1. Pass or Fail

30.0%

Write a Python program that accepts the number of courses and the marks of a student in those courses.  
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

**Input Format:**

- The first input will be an integer  $n$ , the number of courses.
- The second input will be  $n$  integers representing the marks of the student in each of the  $n$  courses, separated by a space.

**Output Format:**  
If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

  
If the student fails any course (marks < 40 in any course), print:

- "Fail".

passorFa...

Submit

1  
2  
3  
4  
5  
6  
7  
8

# Accept number of courses  
n=int(input())  
  
# Accept marks  
marks=list(map(int,input().split()))  
  
# Check fail condition  
if any(m < 40 for m in marks):

Average time  
0.005 s  
6.30 ms

Maximum time  
0.007 s  
7.00 ms

5 out of 5 shown test case(s) passed  
5 out of 5 hidden test case(s) passed

Test case 1  
Expected output  
4  
55 65 78 89  
Aggregate Percentage: 71.75  
Grade: First-Division

Actual output  
4  
55 65 78 89  
Aggregate Percentage: 71.75  
Grade: First-Division

Test case 2

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1.2.2. Fibonacci Series

27.5%

Write a Python program that uses recursion to print the first  $n$  terms of the Fibonacci series.  
**Input Format:**

- A single integer  $n$  representing the number of terms to generate.

**Output Format:**

- A single line containing the first  $n$  terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

Submit

1  
2  
3  
4  
5  
6  
7  
8

def fibonacci(n):  
 if n<=1:  
 return n  
 else:  
 return fibonacci(n-1)+fibonacci(n-2)  
  
n = int(input())

Average time  
0.009 s  
8.75 ms

Maximum time  
0.023 s  
23.00 ms

2 out of 2 shown test case(s) passed  
2 out of 2 hidden test case(s) passed

Test case 1  
Expected output  
5  
0 1 1 2 3

Actual output  
5  
0 1 1 2 3

Test case 2

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1.2.3. Pattern - 1

65.3%

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.  
**Input Format:**

- The input is an integer, representing the number of rows in the pattern.

**Output Format**

- The output should display the pattern of asterisks (\*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

Submit

1  
2  
3

n = int(input())  
for i in range(1,n+1):  
 print("\*" \* i)

Average time  
0.004 s  
3.83 ms

Maximum time  
0.005 s  
5.00 ms

2 out of 2 shown test case(s) passed  
4 out of 4 hidden test case(s) passed

Test case 1  
Expected output  
5  
\*  
\* \*  
\* \* \*  
\* \* \* \*  
\* \* \* \* \*

Actual output  
5  
\*  
\* \*  
\* \* \*  
\* \* \* \*  
\* \* \* \* \*

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1.2.4. Pattern - 200/45

Write a Python program to print a right-angled triangle pattern of numbers.  
**Input Format:**

- The input is an integer, representing the number of rows in the pattern.

**Output Format:**

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

**Note:**

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 n=int(input())
2 for i in range(1,n+1):
3     for j in range(1,i+1):
4         print(j,end=" ")
5     print()
```

Average time0.005 s5.50 msMaximum time0.007 s7.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 16 msDebug

| Expected output | Actual output |
|-----------------|---------------|
| 5               | 5             |
| 1 2             | 1 2           |
| 1 2 3           | 1 2 3         |
| 1 2 3 4         | 1 2 3 4       |
| 1 2 3 4 5       | 1 2 3 4 5     |

TerminalTest cases

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